Adaptive Arena

Sprint 1

Meet The Team

Inna Ivashchenko
Project Manager / Developer

Johnhenry Brayman
Lead Game Developer

Shariah Allen
Game Developer / AI Engineer

Anthony Piombino
Project Manager / QA Engineer

Gabrielle Schwan
UI/UX Developer

Agenda

- Improvements made
- Project Description
- Team Working Agreement
- User Personas
- MVP / Tech / Algorithms
- Diagrams
- Product Backlog / User Stories
- Team Velocity / Burndown
- Demo

Improvements made from Professor Feedback

- More in depth sprint retrospective and sprint planning videos

- Encourage team brainstorming for stronger analysis

Project Description

- 2D-fighting game focusing primarily on AI opponents and improving your general fighting game skills.
- AI will scale appropriately based on selected skill level
  - Easy : Mostly random or suboptimal moves
  - Medium : Basic strategy with a little randomness
  - Hard : Uses machine learning to adapt and respond
- Provide user-friendly interface for user entry
- Tally up and output results of game

Team Working Agreement

Communication And Responsibilities:

- Communication through WhatsApp, Slack, Zoom.
- All team members must respect each other and have the right to contribute their ideas to the project.
- All ideas, vision and changes discussed together, and final decisions are up to most of the team, except urgent situations.
- Each person works within the boundaries established before the start of work.
- Everyone is responsible for their own part.
- All conflicts should first be resolved between the people involved. If that is not possible the issue should be brought to the Team Lead.
- Each member must inform the team about any changes or improvements they make.

Meetings:

- Meeting happens through Zoom, Slack
- The team must have Scrum meetings two times in a week and check WhatsApp, Slack daily to avoid missing any project information.
- Schedule of a Scrum meeting : Tuesday at 8pm and Friday at 9.15pm
- Each team member should participate in at least one meeting per week unless they have an important reason.

User Personas: The Competitive Gamer

Name: Anna Grace
Age: 28
Gender: Female
Occupation: Game Developer & Twitch Streamer
Anna is a season game developer with a solid fan base, that enjoys playing games in her free time, specifically fighting, first-person shooters (FPS), and racing games. She's often searching for new and exciting games that have varying difficult levels and unique features that keeps players engaged.

Challenges:

- Tired of playing the same popular titles, with reoccurring characters and stories
- Wants to find new games developed by small teams to stream for her community

Goals:

- She enjoys playing games and learning different mechanics
- Wants to explore new games with AI elements
- Seeks a game that can be replayed several times

Quote:

" I just want to entertain my community and learn new skills; I love a good challenge!"

User Personas: The Button Smasher

Name: Jordan Carter
Age: 19
Gender: Male
Occupation: Undergraduate Student and Student Athlete

Jordan is a busy student athlete studying animation, that loves to play games on his free time on the weekends or while hanging out with friends. He loves playing games with lots of action, unique characters, and prefers to not memorize combos but just plays for fun, not to win.

Challenges:

- Complex inputs and combinations makes gameplay frustrating
- Long or vague tutorials

Goals:

- Have fun solo or with friends
- Win occasionally, or access to varying difficulties

Quote:

" I skip tutorials, I just want to have mindless fun!"

User Personas: The Lore Explorer

Name: Dylan Taylor
Age: 12
Gender: Male
Occupation: Middle School Student

Dylan enjoys playing games after school with friends, family, or by himself, and specifically enjoys games that are action-packed. He seeks out games that provide a good challenge, has lots of enemies, and visual stimulation. He aspires to be a game developer and enjoys learning how to make 2D games during his video game club meetings.

Challenges:
- Hard to find action games that are age appropriate
- Wants games with unique enemies
- Prefers 2D games

Goals:
- Wants to explore new action games
- Age-appropriate game with visual stimulation

Quote:

" Games are fun when they have a lot of action!"

Inputs

Horizontal Movement:

- Move Forward: Player advances toward the opponent in the right direction using the "D" or "Right" arrow keys

- More Backward: Player retreats from opponent towards the left direction using the "A" or "Left" arrow keys

Vertical Movement:

- Player can do a single or double jump mid-air to evade their opponent's attacks
- Uses the spacebar to do single or double jumps
  - Double tap the spacebar to do a double jump

Blocking:

- Defensive action intended to reduce or prevent incoming damage from attacks
- Use the Right Mouse Button (RMB) to block attacks

Attacks: Punch

- Basic melee attack that deals damage when the player's hitbox collides with the opponent
- Use the Left Mouse Button (LMB) to throw a punch

Reactive Actions

Being Hit

- Triggered once an attack connects.
- Character takes damage
- May briefly enter a hit reaction state once hit
  - Each attack deals varying levels of damage
    - Ex. Punch deals 10 damage

Dying

- Occurs once a fighter's health reaches 0
- Ends Match
- Game transitions to End-Game State

Technologies

Unity 6.3 LTS

Git LFS Extension

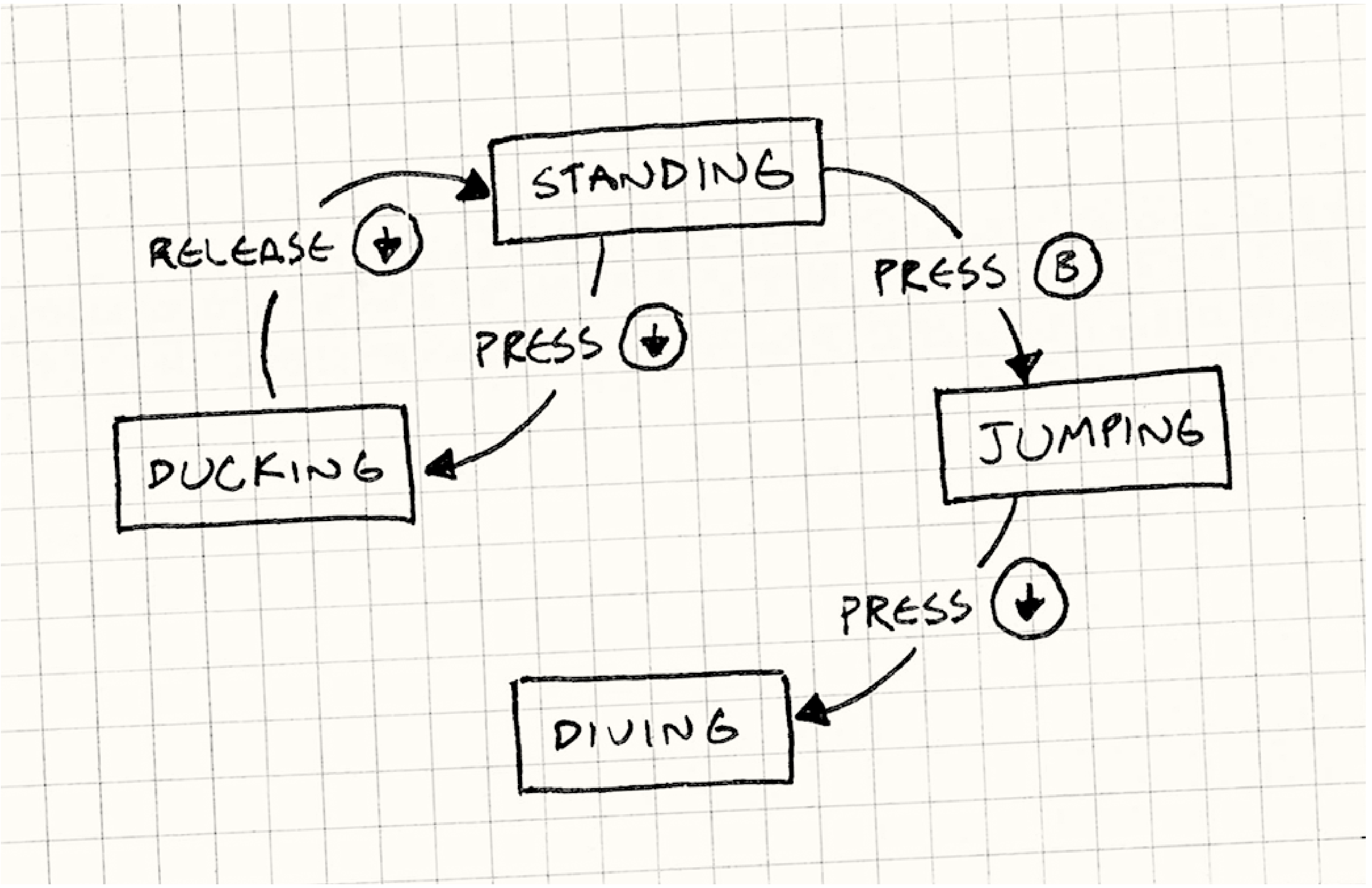
C# Programming

Python (for Machine Learning)

Algorithms

Easy Difficulty : Randomized Finite State Machine

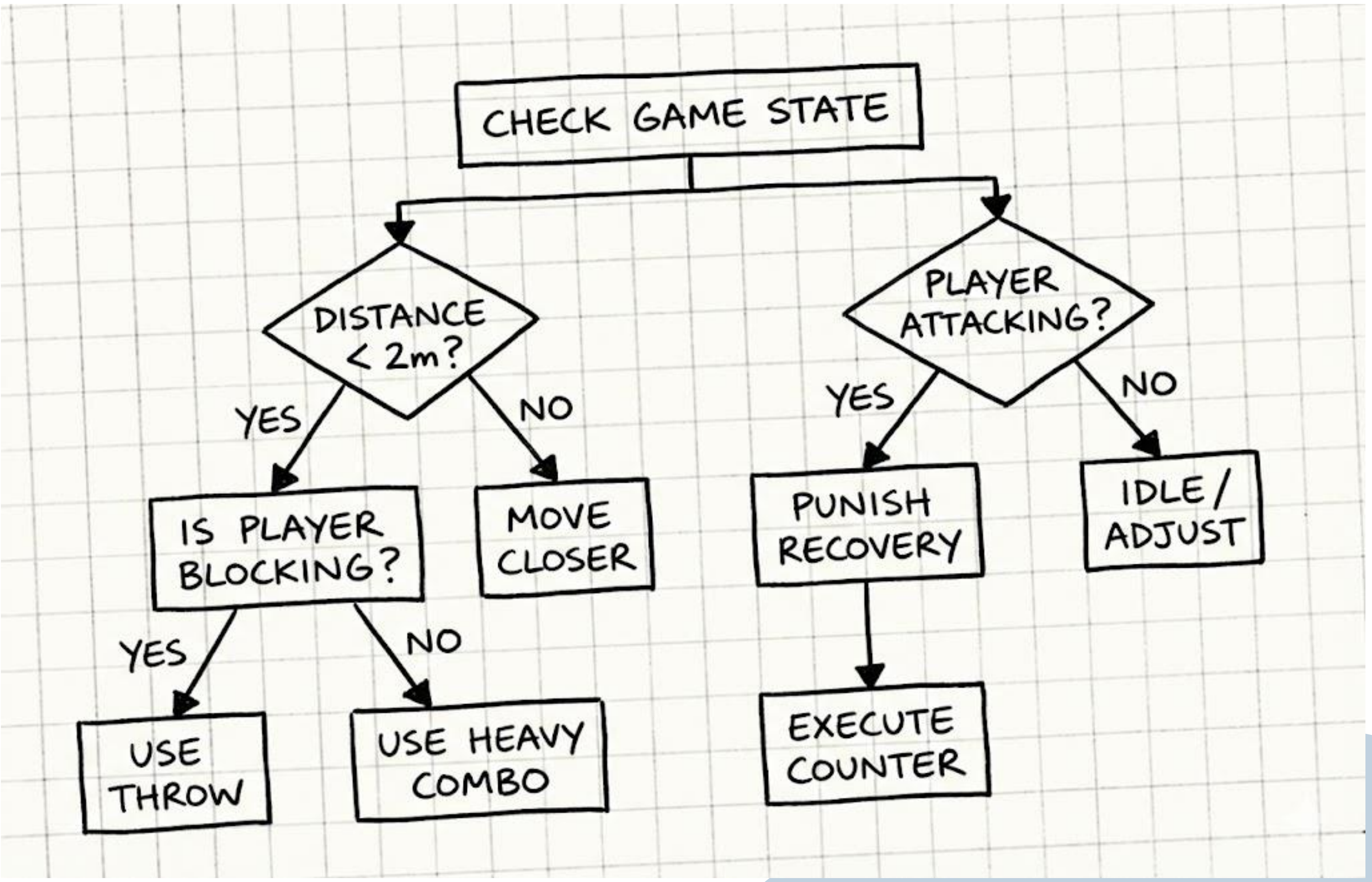
- The enemy AI is presented with multiple pre-defined states (Idle, Attack, Advance, Block, Duck).
- Instead of using strategy, it picks which state to move to next randomly.
- Attacks and movements still make sense but are suboptimal.



Algorithms

Medium Difficulty : Decision Trees

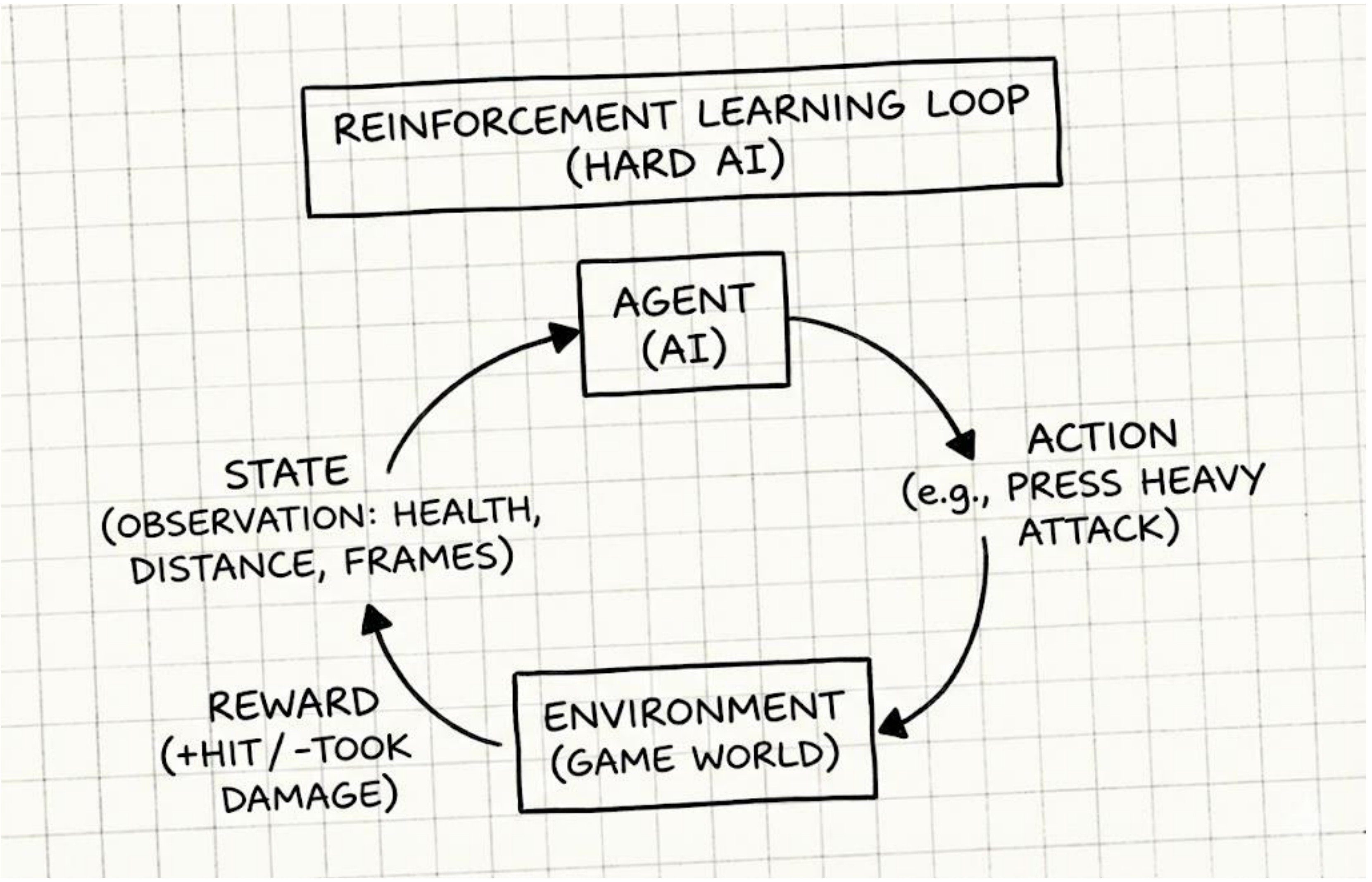
- The enemy AI in this difficulty will start to react differently depending on certain criteria.
- Before making an action, it will check a specific condition and act accordingly.
- Can still make random decisions but in a more educated way.
    - Ex. Check to see if the player is attacking. If yes, randomly choose between block or counter.



Algorithms

Hard Difficulty : Reinforcement Learning

- The enemy AI will fight through many simulations either against itself, or one of the before mentioned difficulties.
- It will receive a reward (+1) for landing a hit on the player, and a punishment (-1) for being struck by the player.
- The AI will learn which actions lead to rewards and which actions lead to punishments.



Minimum Viable Product（MVP)

The Environment:

- A simple flat ground with invisible walls on each side to prevent falling off.
- Simple environment design keeps focus on combat mechanics and player interaction
- A basic camera with a script that keeps both players in-frame while zooming in as close as possible.

The AI Difficulties:

- For the rules-based difficulties, the State Machine and Decision Tree logic must work, even if they are not complex.

- For the hard difficulty, the Unity Agent must be able to read observations and output actions from the Neural Net.

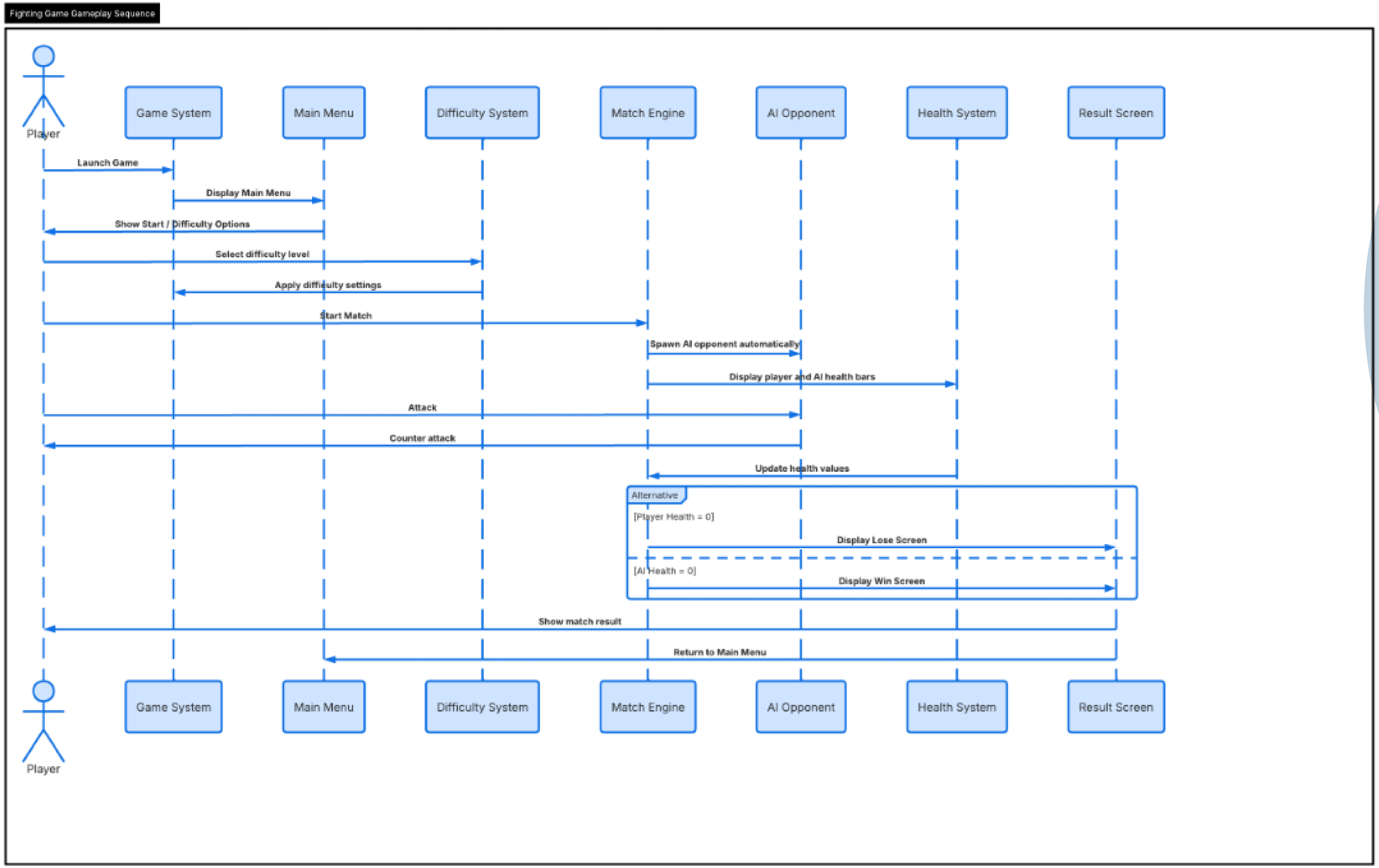
The Combat System:

- The bare minimum controls will be Locomotion, an Attack action, and a Block action.
- Hit detection uses colliders to determine when attacks connect
- Possibility to include combos, ducking, countering, and throwing later.

The Game Flow:

- The player must be able to enter the game from a main menu and select an AI difficulty
- Loads varying end-game scenes depending on match outcome
- In-game must show health bars depleting, and a return to the main menu when one character reaches zero.

Sequence Diagram



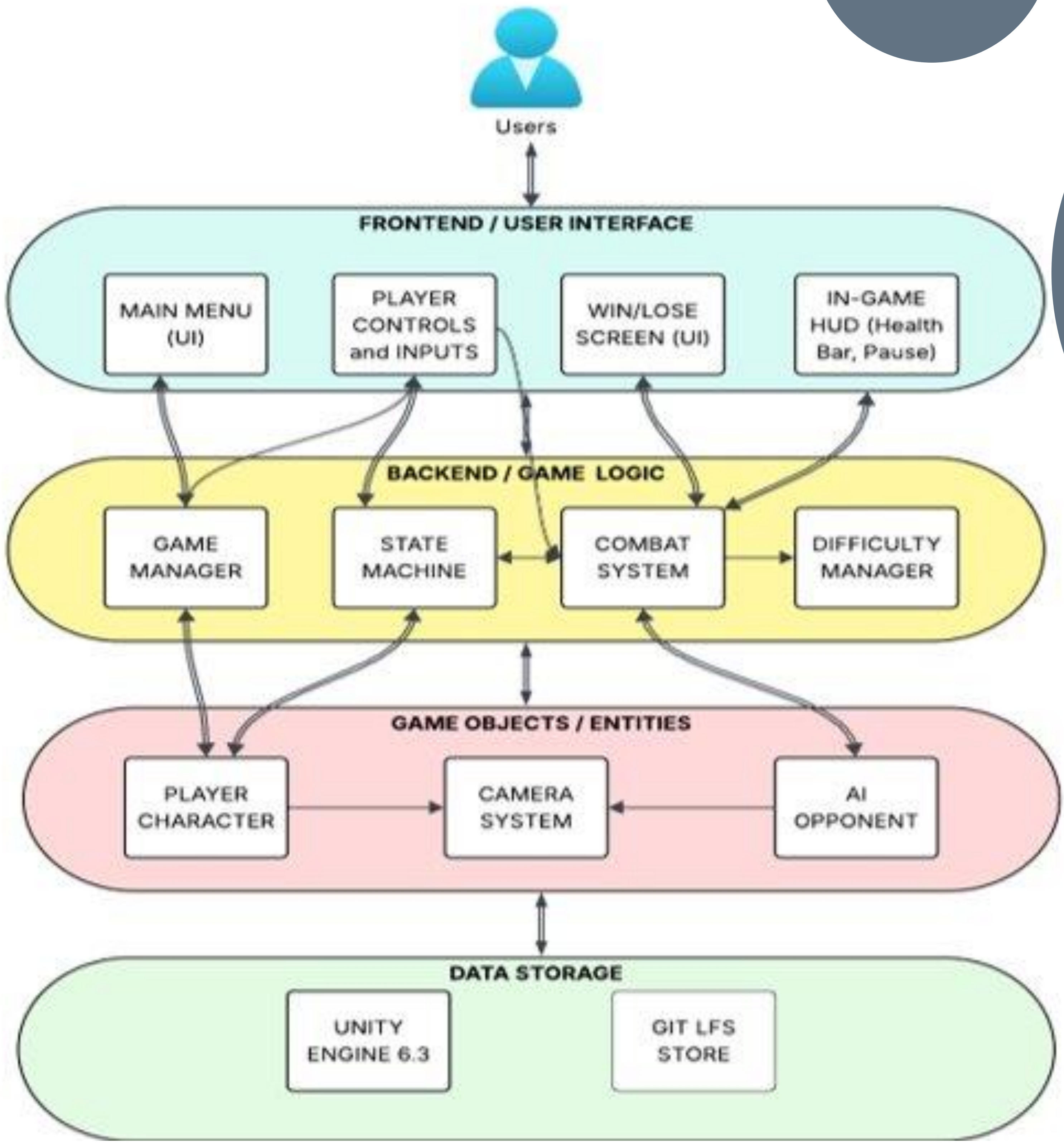
Conceptual Architecture Diagram

Frontend: Accepts player input and displays health status (HUD), Win/Lose Screen.

Backend: The "brain" of the game, managing combat logic, character states, and difficulty.

Entities: Game characters and a "smart" camera, operating according to the backend's rules.

Data Storage: A stable foundation built on Unity 6.3 and Git LFS storage.



Product Backlog

No.	User Stories	Acceptance Criteria	Feature
US_1	As a player, I want to access a main menu so that I can make choices in the game.	Main menu displays on game launch. Menu includes: Start, Settings, Quit, Credits. User can navigate menu using mouse. Selecting Start transitions to fight screen. Selecting Exit closes the game	Game Navigation
US_2	As a player, I want to design my character so that I can play with a customized fighter.	Player can choose character name. Player can design character appearance. Character preview is displayed before confirmation. User confirms selection to proceed. Selected character loads into the match	Character System
US_3	As a player, I want to fight against an AI opponent.	An AI opponent spawns automatically when the match begins. AI character has a visible health bar. AI can move and perform at least one attack. AI attacks reduce the player's health when contact is made.	Combat System
US_4	As a player, I want to select a difficulty level so that the match challenge matches my skill level.	Player can select from at least three difficulty levels: Easy, Medium, and Hard. Difficulty selection occurs before the match begins. Selected difficulty is stored and applied during gameplay. The chosen difficulty level affects AI behavior.	Game Settings
US_5	As a player, I want the AI difficulty to scale based on my selection so that harder modes feel more challenging.	On Easy difficulty, AI attacks less frequently and deals reduced damage. On Medium difficulty, AI has moderate attack frequency and standard damage. On Hard difficulty, AI attacks more frequently and deals increased damage. Differences in difficulty are noticeable during gameplay.	Combat System
US_6	As a player, I want the player and AI characters to use pixel art sprites so that the characters match the retro visual style.	Player character uses a pixel art sprite. AI opponent uses a pixel art sprite. Sprites replace any placeholder shapes for characters and display correctly during gameplay.	Visual Design
US_7	As a player, I want to see a health bar during the fight so that I can track progress.	Player health bar is visible on screen during the match. AI health bar is also visible. Health bars decrease when damage is taken. Health bars update in real time during combat.	Combat UI
US_8	As a player, I want to win or lose based on health depletion so that the fight has a clear outcome.	When the AI's health reaches zero, the player wins. When the player's health reaches zero, the player loses. The match ends immediately when one health bar reaches zero. No further movement or attacks occur after the match ends.	Combat System
US_9	As a player, I want to see a win or lose screen so that I know the match outcome.	A win screen displays when the player wins. A lose screen displays when the player loses. End screen includes options to return to the main menu or play again. Selecting an option transitions properly without crashing the game.	Game Navigation
US_10	As a player, I want smooth transitions between game stages so that gameplay feels complete and polished.	Game transitions smoothly from main menu to character creation, then to difficulty selection, and then to the match. Selected character and difficulty settings carry into the fight. After the match ends, the player can return to the main menu or restart the game without errors or crashes.	Game Architecture
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	Camera System
US_12	As a player, I want the game environment to use pixel art graphics so that the stage matches the retro aesthetic.	The game background uses pixel art design. Background assets load correctly during gameplay. The environment visuals are consistent with the pixel art style and no placeholder backgrounds appear in the Sprint demo.	Visual Design

Product Backlog Continued

US_13	As a player, I want character sprites to animate during movement and attacks so that the game feels dynamic and visually engaging.	Player sprites animate during movement and attacks. AI sprites animate during movement or attacks. Animations display smoothly during gameplay.	Visual Design
US_14	As a player, I want the ability to block attacks so that I can defend myself against enemy attacks during combat.	The player can activate a block action using a designated input. When blocking, incoming attacks deal reduced or no damage. A blocking animation or visual indicator appears while the player is blocking. Blocking only works while the block input is active.	Gameplay
US_15	As a player, I want the game to prevent my character from falling out of the environment so that gameplay stays within the intended boundaries.	The player character cannot move beyond the stage boundaries. The player cannot fall through the ground or platforms. Collision detection prevents the player from leaving the playable area. The player remains within the environment during gameplay.	Gameplay
US_16	As a player, I want the ability to perform a double jump so that I can reach higher areas and have greater mobility during gameplay.	The player can jump once from the ground and perform a second jump while in the air. The second jump can only occur after the first jump. The player cannot perform more than two jumps before landing. The double jump resets after the player lands on the ground.	Gameplay
US_17	As a player, I want to be able to use multiple control methods such as a gamepad controller or keyboard and mouse so that I can play the game using my preferred input device.	The game accepts input from a keyboard and mouse. The game accepts input from a gamepad controller. Core gameplay actions such as movement, jumping, attacking, and blocking can be performed using either control method. The game responds correctly to inputs from either device during gameplay.	Gameplay

Sprint 1 Stories

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint	Team Member
US_1	As a player, I want to access a main menu so that I can make choices in the game.	Main menu displays on game launch. Menu includes: Start, Settings, Quit, Credits. User can navigate menu using mouse. Selecting Start transitions to fight screen. Selecting Exit closes the game	Game Navigation	5	1	1	Gabrielle
US_7	As a player, I want to see a health bar during the fight so that I can track progress.	Player health bar is visible on screen during the match. AI health bar is also visible. Health bars decrease when damage is taken. Health bars update in real time during combat.	Combat UI	5	2	1	Shariah
US_8	As a player, I want to win or lose based on health depletion so that the fight has a clear outcome.	When the AI's health reaches zero, the player wins. When the player's health reaches zero, the player loses. The match ends immediately when one health bar reaches zero. No further movement or attacks occur after the match ends.	Combat System	5	1	1	Johnny
US_9	As a player, I want to see a win or lose screen so that I know the match outcome.	A win screen displays when the player wins. A lose screen displays when the player loses. End screen includes options to return to the main menu or play again. Selecting an option transitions properly without crashing the game.	Game Navigation	3	2	1	Gabrielle
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	Camera System	8	2	1	Shariah

Sprint 1 Stories Continued

US_14	As a player, I want the ability to block attacks so that I can defend myself against enemy attacks during combat.	The player can activate a block action using a designated input. When blocking, incoming attacks deal reduced or no damage. A blocking animation or visual indicator appears while the player is blocking. Blocking only works while the block input is active.	Gameplay	3	3	1	Johnny
US_15	As a player, I want the game to prevent my character from falling out of the environment so that gameplay stays within the intended boundaries.	The player character cannot move beyond the stage boundaries. The player cannot fall through the ground or platforms. Collision detection prevents the player from leaving the playable area. The player remains within the environment during gameplay.	Gameplay	3	2	1	Shariah
US_16	As a player, I want the ability to perform a double jump so that I can reach higher areas and have greater mobility during gameplay.	The player can jump once from the ground and perform a second jump while in the air. The second jump can only occur after the first jump. The player cannot perform more than two jumps before landing. The double jump resets after the player lands on the ground.	Gameplay	6	3	1	Johnny
US_17	As a player, I want to be able to use multiple control methods such as a gamepad controller or keyboard and mouse so that I can play the game using my preferred input device.	The game accepts input from a keyboard and mouse. The game accepts input from a gamepad controller. Core gameplay actions such as movement, jumping, attacking, and blocking can be performed using either control method. The game responds correctly to inputs from either device during gameplay.	Gameplay	5	3	1	Johnny

Sprint 1 Test Cases

No.	User Stories	Acceptance Criteria	Story Points	Total Points Earned			
US_01	As a player, I want to access a main menu so that I can make choices in the game.	Main menu displays on game launch. Menu includes: Start, Settings, Quit, Credits. User can navigate menu using mouse. Selecting Start transitions to fight screen. Selecting Exit closes the game	5	3			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_01_01	Game Launch Displays Main Menu	Launch the game application.	Main menu loads automatically and displays Start Game and Exit options.	1	COMPLETE		
TC_01_02	Play Game	Launch the game and select Start	Game transitions to fight.	1	COMPLETE		
TC_01_03	Exit Game	Launch the game and select Exit from the menu.	The game application closes completely.	1	COMPLETE		
TC_01_03	Credits	Launch the game and select Credits from the menu.	Credits display	1	INCOMPLETE		
TC_01_03	Settings	Launch the game and select Settings from the menu.	Settings display	1	INCOMPLETE		
US_07	As a player, I want to see a health bar during the fight so that I can track progress.	Player health bar is visible on screen during the match. AI health bar is also visible. Health bars decrease when damage is taken. Health bars update in real time during combat.	5	5			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_07_01	Health Bars Visible	Start a match and observe the game interface.	Player and AI health bars are visible on the screen.	1	COMPLETE		
TC_07_02	Health Bar Updates	Start a match and perform attacks between player and AI.	Health bars decrease appropriately when damage occurs.	2	COMPLETE		
TC_07_03	Real Time Health Updates	Continue attacking during combat.	Health bars update immediately as damage is applied.	2	COMPLETE		
US_08	As a player, I want to win or lose based on health depletion so that the fight has a clear outcome.	When the AI's health reaches zero, the player wins. When the player's health reaches zero, the player loses. The match ends immediately when one health bar reaches zero. No further movement or attacks occur after the match ends.	5	4			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_08_01	Player Win Condition	Start a match and reduce AI health to zero.	Victory screen appears and match ends.	1	COMPLETE		
TC_08_02	Player Lose Condition	Start a match and allow AI to reduce player health to zero.	Lose screen appears and match ends.	1	INCOMPLETE		
TC_08_03	Combat Stops After Match End	Reach a win or loss condition.	No further attacks or movement occur after the match ends.	3	COMPLETE		
US_09	As a player, I want to see a win or lose screen so that I know the match outcome.	A win screen displays when the player wins. A lose screen displays when the player loses. End screen includes options to return to the main menu or play again. Selecting an option transitions properly without crashing the game.	3	3			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_09_01	Win Screen Display	Win a match by reducing AI health to zero.	Victory screen appears showing victory result.	1	COMPLETE		
TC_09_02	Lose Screen Display	Lose a match by reaching zero health.	Lose screen appears showing defeat result.	1	COMPLETE		
TC_09_03	End Screen Options	From win or lose screen select Retry or Main Menu.	Selected option transitions correctly without crashing.	1	COMPLETE		

Sprint 1 Test Cases Continued

US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	8	2			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_11_01	Camera Shows Both Fighters	Start a match and observe camera positioning during gameplay.	Both fighters remain visible on screen at the same time.	3	INCOMPLETE		
TC_11_02	Camera Follows Fighter Movement	Move fighters across the stage to opposite ends.	Camera adjusts position to keep both fighters visible.	3	INCOMPLETE		
TC_11_03	Camera Movement Smoothness	Move fighters quickly across the stage while fighting.	Camera movement remains smooth and does not abruptly jump or lose view of fighters.	2	COMPLETE		
US_14	As a player, I want the ability to block attacks so that I can defend myself against enemy attacks during combat.	The player can activate a block action using a designated input. When blocking, incoming attacks deal reduced or no damage. A blocking animation or visual indicator appears while the player is blocking. Blocking only works while the block input is active.	3	2			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_14_01	Player Can Activate Block	Launch the game and start a match. Press the designated block input during gameplay.	The player character enters a blocking state or displays a blocking animation or indicator.	1	COMPLETE		
TC_14_02	Block Reduces or Prevents Damage	Start a match and activate the block input. Allow the AI opponent to perform an attack while the player is blocking.	The player receives reduced damage while the block action is active.	1	INCOMPLETE		
TC_14_03	Block Only Works While Input Is Active	Start a match and activate the block input, then release the block input and allow the AI opponent to attack.	Blocking only occurs while the block input is held. When released, the player no longer blocks attacks.	1	COMPLETE		
US_15	As a player, I want the game to prevent my character from falling out of the environment so that gameplay stays within the intended boundaries.	The player character cannot move beyond the stage boundaries. The player cannot fall through the ground or platforms. Collision detection prevents the player from leaving the playable area. The player remains within the environment during gameplay.	3	3			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_15_01	Player Cannot Move Beyond Stage Boundary	Launch the game and start a match. Move the player character toward the left or right edge of the stage. Continue moving in that direction.	The player stops at the stage boundary and cannot move beyond the playable area.	1	COMPLETE		
TC_15_02	Player Cannot Fall Through Ground	Start a match and move the player around the stage while observing the ground collision.	The player remains on the ground and does not fall through the environment.	1	COMPLETE		
TC_15_03	Collision Prevents Leaving Playable Area	Start a match and attempt to move the player character toward the edges or corners of the environment	Collision detection prevents the player from leaving the environment or moving outside the playable area.	1	COMPLETE		

Sprint 1 Test Cases Continued

		environment.					
US_16	As a player, I want the ability to perform a double jump so that I can reach higher areas and have greater mobility during gameplay.	The player can jump once from the ground and perform a second jump while in the air. The second jump can only occur after the first jump. The player cannot perform more than two jumps before landing. The double jump resets after the player lands on the ground.	6	6			
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status	Exectution Date	Tester
TC_16_01	Player Can Perform First Jump	Launch the game and start a match. Press the jump input while the player is standing on the ground.	The player character jumps into the air.	1	COMPLETE		
TC_16_02	Player Can Perform Double Jump	Start a match and press the jump input once to jump. While still in the air, press the jump input again.	The player performs a second jump while airborne.	1	COMPLETE		
TC_16_03	Double Jump Limit Enforced	Start a match and press the jump input twice while airborne. Attempt to press the jump input a third time before landing.	The player cannot perform a third jump and must land before jumping again.	2	COMPLETE		
TC_16_04	Double Jump Resets After Landing	Start a match and perform a double jump. Allow the player to land on the ground and then attempt to jump again.	The player can again perform two jumps after landing.	2	COMPLETE		
US_17	As a player, I want to be able to use multiple control methods such as a gamepad controller or keyboard and mouse so that I can play the game using my preferred input device.	The game accepts input from a keyboard and mouse. The game accepts input from a gamepad controller. Core gameplay actions such as movement, jumping, attacking, and blocking can be performed using either control method. The game responds correctly to inputs from either device during gameplay.	3	1			
No.	Test Case Title	Test Flow	Expected Result	Points Breakdown	Execution Status	Exectution Date	Tester
TC_17_01	Keyboard and Mouse Controls Work	Launch the game and start a match. Use the keyboard and mouse to perform actions such as movement, jumping, attacking, and blocking.	The player character responds correctly to all keyboard and mouse inputs.	1	COMPLETE		
TC_17_02	Gamepad Controller Controls Work	Launch the game with a connected gamepad controller and start a match. Use the controller to perform actions such as movement, jumping, attacking, and blocking.	The player character responds correctly to all controller inputs.	1	INCOMPLETE		
TC_17_03	Game Responds to Active Input Device	Launch the game with both a controller and keyboard connected. Start a match and perform actions using each input device.	The game correctly registers inputs from either the keyboard and mouse or the controller during gameplay.	1	INCOMPLETE		

Sprint 1 Stories Completed

No.	User Stories	Acceptance Criteria	Story Points	Total Points Earned			
US_07	As a player, I want to see a health bar during the fight so that I can track progress.	Player health bar is visible on screen during the match. AI health bar is also visible. Health bars decrease when damage is taken. Health bars update in real time during combat.	5	5			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_07_01	Health Bars Visible	Start a match and observe the game interface.	Player and AI health bars are visible on the screen.	1	COMPLETE		
TC_07_02	Health Bar Updates	Start a match and perform attacks between player and AI.	Health bars decrease appropriately when damage occurs.	2	COMPLETE		
TC_07_03	Real Time Health Updates	Continue attacking during combat.	Health bars update immediately as damage is applied.	2	COMPLETE		
US_09	As a player, I want to see a win or lose screen so that I know the match outcome.	A win screen displays when the player wins. A lose screen displays when the player loses. End screen includes options to return to the main menu or play again. Selecting an option transitions properly without crashing the game.	3	3			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_09_01	Win Screen Display	Win a match by reducing AI health to zero.	Victory screen appears showing victory result.	1	COMPLETE		
TC_09_02	Lose Screen Display	Lose a match by reaching zero health.	Lose screen appears showing defeat result.	1	COMPLETE		
TC_09_03	End Screen Options	From win or lose screen select Retry or Main Menu.	Selected option transitions correctly without crashing.	1	COMPLETE		

Sprint 1 Stories Completed

US_15	As a player, I want the game to prevent my character from falling out of the environment so that gameplay stays within the intended boundaries.	The player character cannot move beyond the stage boundaries. The player cannot fall through the ground or platforms. Collision detection prevents the player from leaving the playable area. The player remains within the environment during gameplay.	3	3			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_15_01	Player Cannot Move Beyond Stage Boundary	Launch the game and start a match. Move the player character toward the left or right edge of the stage. Continue moving in that direction.	The player stops at the stage boundary and cannot move beyond the playable area.	1	COMPLETE		
TC_15_02	Player Cannot Fall Through Ground	Start a match and move the player around the stage while observing the ground collision.	The player remains on the ground and does not fall through the environment.	1	COMPLETE		
TC_15_03	Collision Prevents Leaving Playable Area	Start a match and attempt to move the player character toward the edges or corners of the environment.	Collision detection prevents the player from leaving the environment or moving outside the playable area.	1	COMPLETE		
US_16	As a player, I want the ability to perform a double jump so that I can reach higher areas and have greater mobility during gameplay.	The player can jump once from the ground and perform a second jump while in the air. The second jump can only occur after the first jump. The player cannot perform more than two jumps before landing. The double jump resets after the player lands on the ground.	6	6			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_16_01	Player Can Perform First Jump	Launch the game and start a match. Press the jump input while the player is standing on the ground.	The player character jumps into the air.	1	COMPLETE		
TC_16_02	Player Can Perform Double Jump	Start a match and press the jump input once to jump. While still in the air, press the jump input again.	The player performs a second jump while airborne.	1	COMPLETE		
TC_16_03	Double Jump Limit Enforced	Start a match and press the jump input twice while airborne. Attempt to press the jump input a third time before landing.	The player cannot perform a third jump and must land before jumping again.	2	COMPLETE		
TC_16_04	Double Jump Resets After Landing	Start a match and perform a double jump. Allow the player to land on the ground and then attempt to jump again.	The player can again perform two jumps after landing.	2	COMPLETE		

Sprint 1 Stories Not Completed

No.	User Stories	Acceptance Criteria	Story Points	Total Points Earned			
US_01	As a player, I want to access a main menu so that I can make choices in the game.	Main menu displays on game launch. Menu includes: Start, Settings, Quit, Credits. User can navigate menu using mouse. Selecting Start transitions to fight screen. Selecting Exit closes the game	5	3			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_01_01	Game Launch Displays Main Menu	Launch the game application.	Main menu loads automatically and displays Start Game and Exit options.	1	COMPLETE		
TC_01_02	Play Game	Launch the game and select Start	Game transitions to fight.	1	COMPLETE		
TC_01_03	Exit Game	Launch the game and select Exit from the menu.	The game application closes completely.	1	COMPLETE		
TC_01_03	Credits	Launch the game and select Credits from the menu.	Credits display	1	INCOMPLETE		
TC_01_03	Settings	Launch the game and select Settings from the menu.	Settings display	1	INCOMPLETE		
US_08	As a player, I want to win or lose based on health depletion so that the fight has a clear outcome.	When the AI's health reaches zero, the player wins. When the player's health reaches zero, the player loses. The match ends immediately when one health bar reaches zero. No further movement or attacks occur after the match ends.	5	4			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_08_01	Player Win Condition	Start a match and reduce AI health to zero.	Victory screen appears and match ends.	1	COMPLETE		
TC_08_02	Player Lose Condition	Start a match and allow AI to reduce player health to zero.	Lose screen appears and match ends.	1	INCOMPLETE		
TC_08_03	Combat Stops After Match End	Reach a win or loss condition.	No further attacks or movement occur after the match ends.	3	COMPLETE		
US_11	As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent.	The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions.	8	2			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_11_01	Camera Shows Both Fighters	Start a match and observe camera positioning during gameplay.	Both fighters remain visible on screen at the same time.	3	INCOMPLETE		
TC_11_02	Camera Follows Fighter Movement	Move fighters across the stage to opposite ends.	Camera adjusts position to keep both fighters visible.	3	INCOMPLETE		
TC_11_03	Camera Movement Smoothness	Move fighters quickly across the stage while fighting.	Camera movement remains smooth and does not abruptly jump or lose view of fighters.	2	COMPLETE		

Sprint 1 Stories Not Completed

			lose view of fighters.				
US_14	As a player, I want the ability to block attacks so that I can defend myself against enemy attacks during combat.	The player can activate a block action using a designated input. When blocking, incoming attacks deal reduced or no damage. A blocking animation or visual indicator appears while the player is blocking. Blocking only works while the block input is active.	3	2			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_14_01	Player Can Activate Block	Launch the game and start a match. Press the designated block input during gameplay.	The player character enters a blocking state or displays a blocking animation or indicator.	1	COMPLETE		
TC_14_02	Block Reduces or Prevents Damage	Start a match and activate the block input. Allow the AI opponent to perform an attack while the player is blocking.	The player receives reduced damage while the block action is active.	1	INCOMPLETE		
TC_14_03	Block Only Works While Input Is Active	Start a match and activate the block input, then release the block input and allow the AI opponent to attack.	Blocking only occurs while the block input is held. When released, the player no longer blocks attacks.	1	COMPLETE		
US_17	As a player, I want to be able to use multiple control methods such as a gamepad controller or keyboard and mouse so that I can play the game using my preferred input device.	The game accepts input from a keyboard and mouse. The game accepts input from a gamepad controller. Core gameplay actions such as movement, jumping, attacking, and blocking can be performed using either control method. The game responds correctly to inputs from either device during gameplay.	3	1			
No.	**Test Case Title**	**Test Flow**	**Expected Result**	**Points Breakdown**	**Execution Status**	**Exectution Date**	**Tester**
TC_17_01	Keyboard and Mouse Controls Work	Launch the game and start a match. Use the keyboard and mouse to perform actions such as movement, jumping, attacking, and blocking.	The player character responds correctly to all keyboard and mouse inputs.	1	COMPLETE		
TC_17_02	Gamepad Controller Controls Work	Launch the game with a connected gamepad controller and start a match. Use the controller to perform actions such as movement, jumping, attacking, and blocking.	The player character responds correctly to all controller inputs.	1	INCOMPLETE		
TC_17_03	Game Responds to Active Input Device	Launch the game with both a controller and keyboard connected. Start a match and perform actions using each input device.	The game correctly registers inputs from either the keyboard and mouse or the controller during gameplay.	1	INCOMPLETE		

Stories Planned for Sprint 2

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint	Team Member
US_3	As a player, I want to fight against an AI opponent.	An AI opponent spawns automatically when the match begins. AI character has a visible health bar. AI can move and perform at least one attack. AI attacks reduce the player's health when contact is made.	Combat System	8	1	2	Johnny
US_4	As a player, I want to select a difficulty level so that the match challenge matches my skill level.	Player can select from at least three difficulty levels: Easy, Medium, and Hard. Difficulty selection occurs before the match begins. Selected difficulty is stored and applied during gameplay. The chosen difficulty level affects AI behavior.	Game Settings	3	1	2	Johnny
US_5	As a player, I want the AI difficulty to scale based on my selection so that harder modes feel more challenging.	On Easy difficulty, AI attacks less frequently and deals reduced damage. On Medium difficulty, AI has moderate attack frequency and standard damage. On Hard difficulty, AI attacks more frequently and deals increased damage. Differences in difficulty are noticeable during gameplay.	Combat System	5	1	2	Johnny

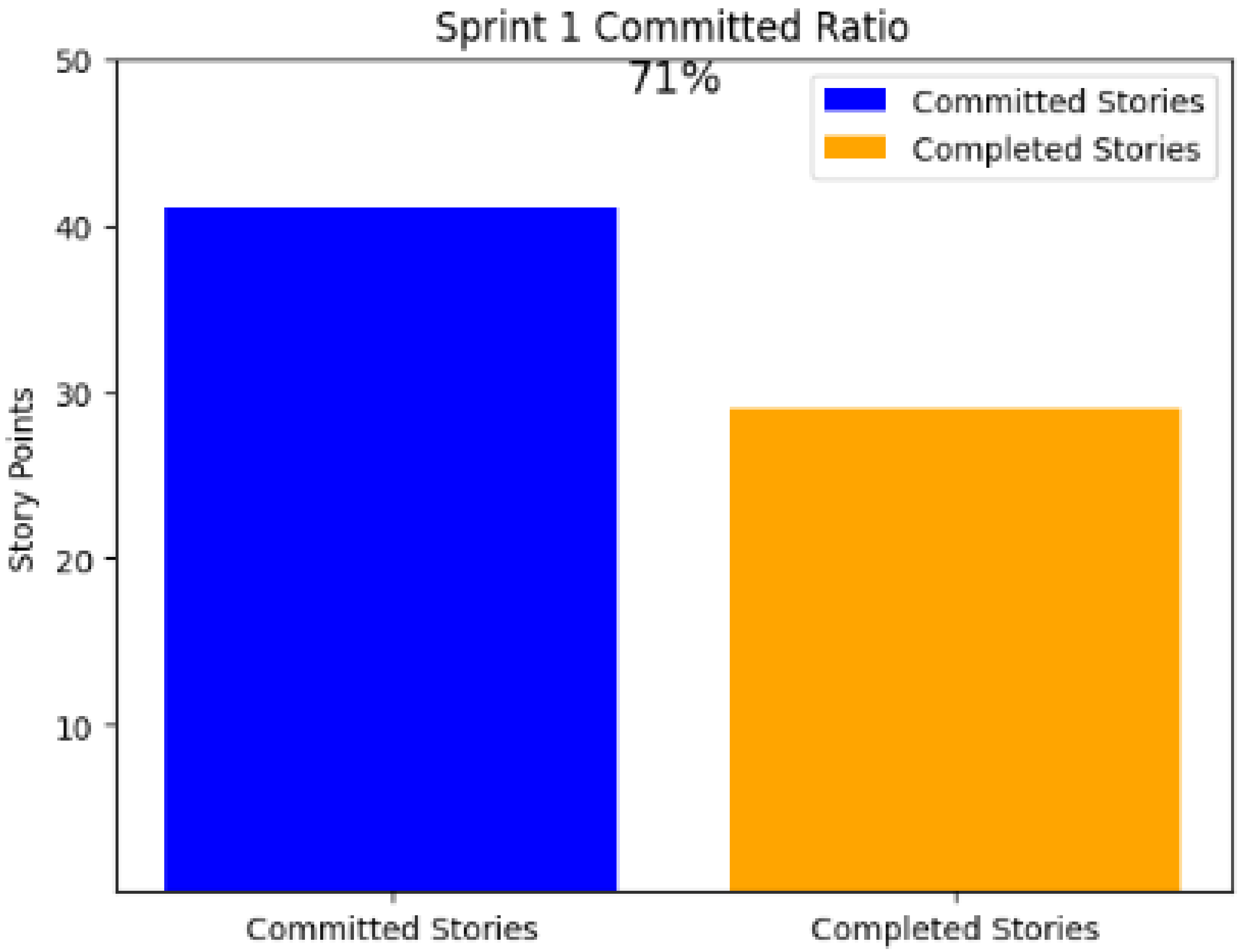
Stories Planned for Sprint 3

No.	User Stories	Acceptance Criteria	Feature	Story Points	Priority	Sprint	Team Member
US_2	As a player, I want to design my character so that I can play with a customized fighter.	Player can choose character name. Player can design character appearance. Character preview is displayed before confirmation. User confirms selection to proceed. Selected character loads into the match	Character System	5	2	3	Johnny
US_6	As a player, I want the player and AI characters to use pixel art sprites so that the characters match the retro visual style.	Player character uses a pixel art sprite. AI opponent uses a pixel art sprite. Sprites replace any placeholder shapes for characters and display correctly during gameplay.	Visual Design	3	3	3	Shariah
US_12	As a player, I want the game environment to use pixel art graphics so that the stage matches the retro aesthetic.	The game background uses pixel art design. Background assets load correctly during gameplay. The environment visuals are consistent with the pixel art style and no placeholder backgrounds appear in the Sprint demo.	Visual Design	3	3	3	Shariah
US_13	As a player, I want character sprites to animate during movement and attacks so that the game feels dynamic and visually engaging.	Player sprites animate during movement and attacks. AI sprites animate during movement or attacks. Animations display smoothly during gameplay.	Visual Design	3	3	3	Shariah

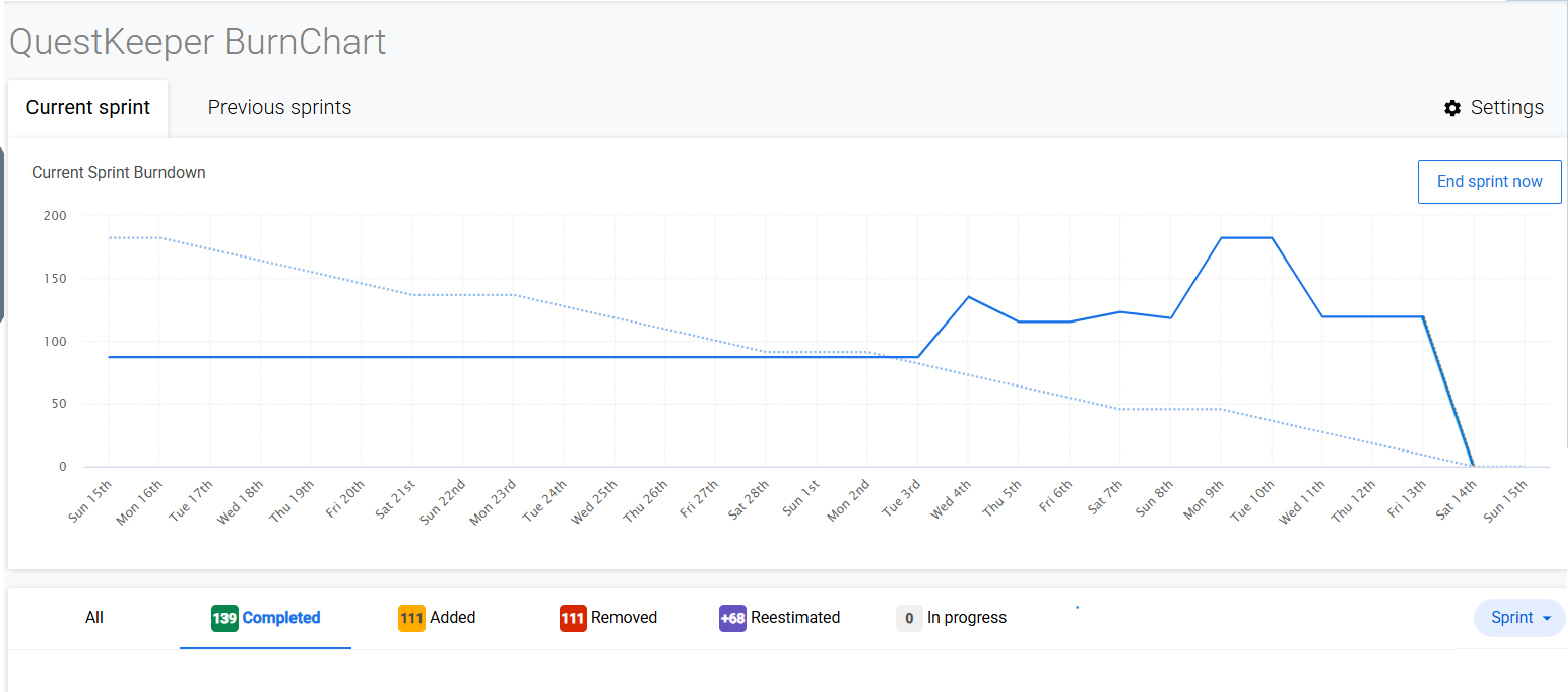
Team Velocity

Story Points Committed: 41

Story Points Completed: 29



Burndown Chart



Retrospective Sprint 1

What went well

- Defined job roles and responsibilities clearly. Great technical skills for unity deliverables. Also great test case document. +0
- good understanding, logic and flow of the game +0
- Better organized with the tasks and timeliness of the deliverables. +0
- Communication improved. Slack was a major help to organize conversations and coordinate with team members. +0
- More organization = Less stress to create a great result...aka a good final product/ game demo. +0
- Good time management and attention to due dates +0

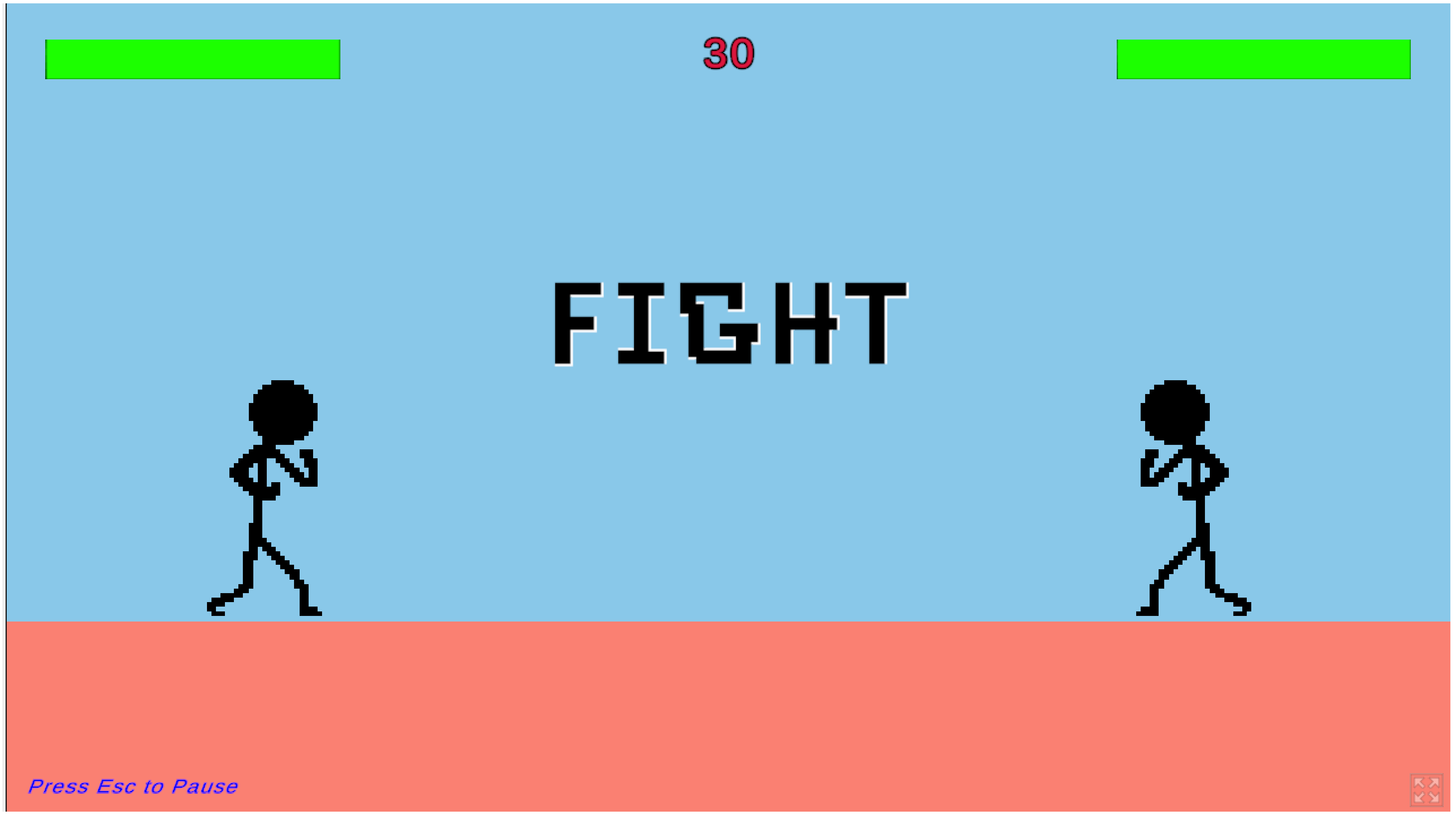
What did not go well

- Trello Review first to increase clarity of tasks assigned to each team member. Tasks were too general/large. +0
- Specify tasks past the required user stories. +0
- Github/unity branch changes and merging them...cleaning +0
- LFS extension download on github caused major errors to the main branch +0
- Moving forward, implementing new features (ai) duplicate main scene to do your work in unity. Easier way to merge work in the end. +0
- test scenes for enemy ai +0

Action items

- Update and clarify tasks on Trello early into the sprint planning to make sure everyone knows their responsibilities. Include details like dates to be finished and specific checklist items. +0
- Lay out individual tasks to create a better work flow. +0
- Work ahead instead of adjacent. In the beginning go through the full list of tasks that need to be finished (prioritize the rubric) +0
- Game is setup and changes should hopefully run smoother +0
- Being cautious of downloads and communicate with the team. Work in your own branch for changes. New branch for new work. +0
- More technical work for all team members...Code review Create slack channel for the comments before approval +0

Project Demo



Adaptive Arena Demo

Wiki Page Link

https://github.com/htmw/2026SA-QuestKeeper/wiki

Thank You